Documenting a network experiment

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make another person able to check the entire result.

Use exact data, checkpoint and prediction rows, with explicit unmeasured fields and hash limitations.

Keep one habit: Model a complete record → diagnose missing evidence → assess fresh R17 → plan a documented replay.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Record a network experiment with enough detail to **replay** its calculations

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

It omits the task, data and splits, model and code versions, quality counts or loss definition, hardware, timing boundary and raw results. Another person cannot check what was measured or repeat the comparison.

Opening retrieval

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate throughput and amortised time per example from repeated timings

Reference example, not the only acceptable wording: Teacher explanation: A: 2.0 seconds, 20 ms per example, 50 examples per second. B: 1.2 seconds, 12 ms per example, about 83.33 examples per second.

The boundary covers 100 completed sequential **prediction**s on pre**process**ed **input**s, excluding loading and pre**process**ing. It does not establish worst-case or percentile request latency.

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a predicted class, confidence score and observed correctness

Reference example, not the only acceptable wording: Teacher explanation: B predicts 1 with confidence 0.90 and is wrong. E predicts 0 with confidence 0.55 and is correct. F predicts 0 with confidence 0.90 and is wrong.

F's low score for class 1 corresponds to a high score for class 0, the class selected.

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

Look for: Look up a vector using a token ID and preserve sequence shape

Reference example, not the only acceptable wording: Teacher explanation: Vectors are [[1, 1], [0, 1], [0, 2], [0, 0]], mask [1, 1, 1, 0] and shape 4 by 2.

Pictured end goal

By the end, I can explain documenting a network experiment using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Efficiency and resource measurement

Uncertainty and failure analysis

Embeddings and neighbourhoods

For the compact alternative schedule, use two 45-minute sessions. Treat the tiny ReLU model as a fully inspectable experiment, not a claim about real-world accuracy. The code trains one scalar on two rows and selects using one validation row.

Require students to label metrics, denominators and data roles before judging the result; no final evaluation or resource measurement is supplied.

Outcomes

Record a network experiment with enough detail to **replay** its calculations

Trace reported metrics to data roles and saved parameters

Write a bounded conclusion and identify missing evaluation **evidence**

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach a continuous chain from question to bounded conclusion

> Keep training gradients, validation selection and missing final evidence distinct.

Question, settings and complete data manifest

A useful experiment record explains what was asked, what was done, what happened and what remains unknown.

Use run NN32-A: does **fitting** one positive ReLU weight reduce validation mean squared error compared with its initial value?

This tiny constructed regression task teaches record keeping; it cannot establish performance on real-world data.

The network predicts max(0, a*x), with one weight a and no bias. Start at a = 1, use learning rate 0.1, two full-batch **training** updates and mean squared error, the mean of squared **prediction** errors.

There is no normalisation, random initialisation or shuffling. Use Python float arithmetic and record the actual Python version when **replay**ing.

Before running, set the selection rule: choose the **checkpoint** with the smaller validation MSE, breaking a tie toward the earlier update. The initial model is the baseline.

Validation V1 chooses a **checkpoint** but never contributes to the **training** gradient. Final evaluation has not been run; its result must be recorded as not measured.

Data role and ID	Input x	Target y
Training T1	1	2
Training T2	2	4
Validation V1	3	6

Checked gradients and complete metric ledger

For these two **training** cases, MSE = 2.5*(a - 2) squared and its gradient is 5*(a - 2). At a = 1 the gradient is -5, so the first update is 1 - 0.1*(-5) = 1.5. The next gradient is -2.5, giving a = 1.75.

All preactivations in this run stay positive, so ReLU’s derivative is one; this is not a general backpropagation implementation.

The code assumes the supplied nonempty numeric datasets and positive preactivations. Each record stores update number, weight, **training** MSE and validation MSE after the update.

Keep both records, the initial weight and the rule; saving only the best score erases how the choice was made.

State · Gradient before update · Weight · Train predictions T1/T2 · Training MSE · V1 prediction · Validation MSE

> **State:** Initial baseline
> **Gradient before update:** Not yet updated
> **Weight:** 1
> **Train predictions T1/T2:** 1 / 2
> **Training MSE:** 2.5
> **V1 prediction:** 3
> **Validation MSE:** 9

> **State:** Update 1
> **Gradient before update:** -5
> **Weight:** 1.5
> **Train predictions T1/T2:** 1.5 / 3
> **Training MSE:** 0.625
> **V1 prediction:** 4.5
> **Validation MSE:** 2.25

> **State:** Update 2
> **Gradient before update:** -2.5
> **Weight:** 1.75
> **Train predictions T1/T2:** 1.75 / 3.5
> **Training MSE:** 0.15625
> **V1 prediction:** 5.25

> **Validation MSE:** 0.5625

Main prediction key and bounded conclusion

Read each checkpoint against the same validation case

CHECKPOINT WEIGHT	TRAINING MSE	VALIDATION MSE
Update 1: a = 1.5	0.625	2.25
Update 2: a = 1.75	0.15625	0.5625

The baseline validation MSE is 9. The predeclared rule selects update 2 with validation MSE 0.5625 on one validation case. This is not a final-test score.

Guided: 1.75*3 = 5.25, error 5.25 - 6 = -0.75, squared error 0.5625. V1 selected the **checkpoint** and is validation, not untouched final evaluation.

Exit: replace final accuracy with validation MSE 0.5625 on one constructed case selected using that case. Energy is not measured.

Supply data and splits, exact code, settings, baseline and both **checkpoint**s, raw **prediction**s, metric definition, selection rule and environment information.

A defensible conclusion is: in run NN32-A, update 2 reduced MSE on the single constructed validation case from 9 to 0.5625 compared with the initial model.

This **checkpoint** was selected using that same validation case; generalisation is untested. No timing, peak-memory or energy measurements are provided.

> **Using the selected weight, calculate the prediction for V1 and its squared error. Explain why the result must be labelled validation rather than independent final evaluation.**

Artifact record and hash limitations

Artifact or record	Why another person needs it
Data manifest: T1/T2 training, V1 validation	Recover exact inputs, targets, IDs, roles and ordering
Exact code and configuration	Recover model formula, loss, rate, initial state, update count and assumptions
Baseline and every checkpoint	Reconstruct the comparison and selection history
Raw predictions and metric calculations	Trace each reported squared error to a case and weight
Python/environment versions and actual run status	Separate a real replay from a paper trace and record possible differences
Content hashes for exact saved versions	Detect changed content; do not prove reference correctness or study quality
Final evaluation, time, peak memory, energy	Record not measured when absent; do not replace with zero
Changed data or preprocessing	Preserve original artifacts and record a revised run rather than overwrite history

Separate pictured run and scope

Document **checkpoint** selection so another team can reconstruct every reported number.

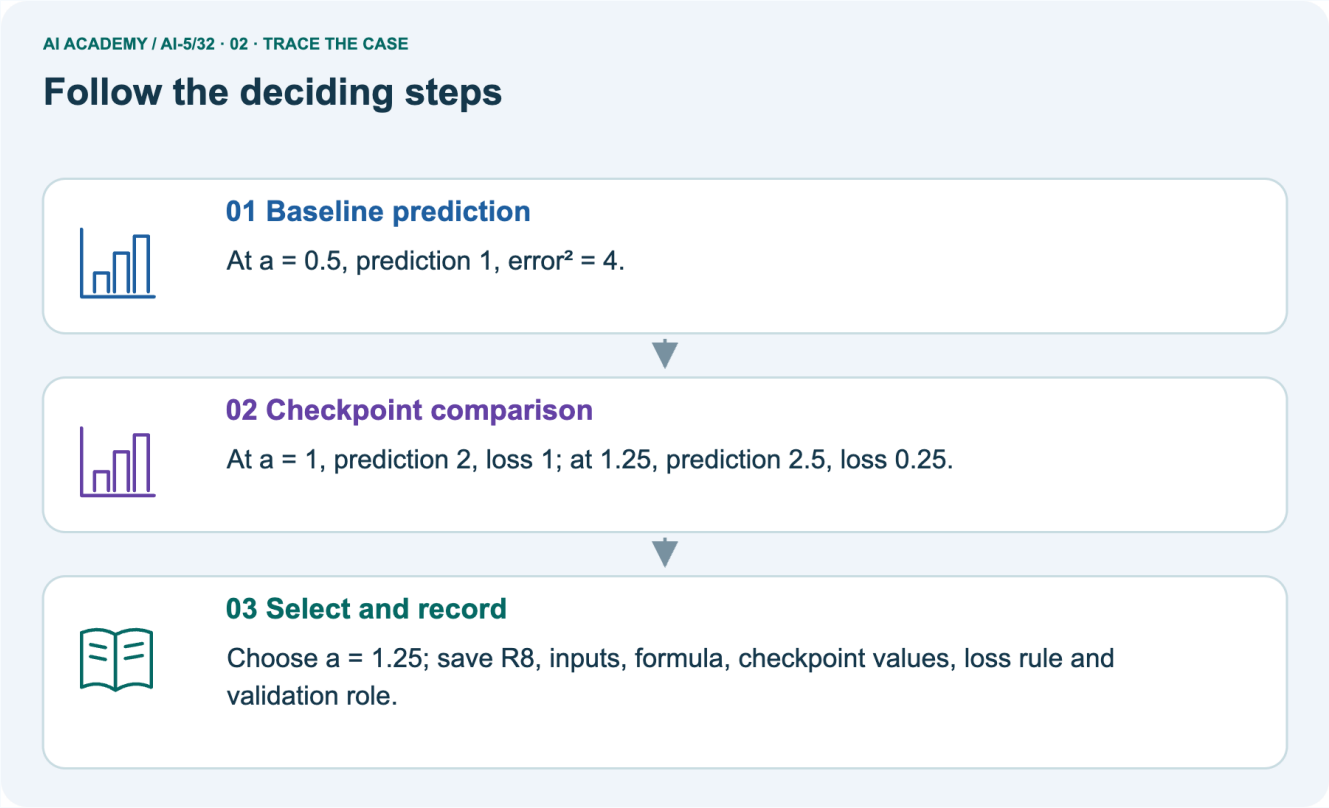
Run R8 predicts ReLU(a×x). Validation V4 has x = 2, target 3. Baseline a = 0.5; **checkpoint**s a = 1 and 1.25; choose minimum validation squared error.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main NN32-A uses V1 x = 3, target 6: baseline validation MSE 9, then 2.25 and .5625. The picture is a separate run R8 using V4 x = 2, target 3: supplied weights .5, 1, 1.25 give predictions 1, 2, 2.5 and squared errors 4, 1, .25. It selects weight 1.25, but no training

procedure for those pictured checkpoints is supplied. Both examples use validation to choose; neither provides independent final evaluation.

Picture trace and validation-renaming key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Baseline prediction	At a = 0.5, prediction 1, error² = 4.
2. Checkpoint comparison	At a = 1, prediction 2, loss 1; at 1.25, prediction 2.5, loss 0.25.
3. Select and record	Choose a = 1.25; save R8, inputs, formula, checkpoint values, loss rule and validation role.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A reproducible record includes the actual **input**, formula, parameters and scoring rule, together with each dataset's role. The validation case selects the best supplied **checkpoint**, so it is part of development.

Save that selection history and freeze the **checkpoint** before evaluating fresh final cases. Renaming the same case a test does not change how it was used.

Reconstructing one run verifies arithmetic and provenance, not population performance or educational learning impact.

Change: A teammate calls V4 a final test after using it to select a=1.25.

Prediction: Correct its role to validation; it was used for selection.

Reason: Changing the label on a dataset does not make its **evidence** untouched.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Run identifier	A unique label linking one experiment’s settings, outputs and records
Checkpoint	Saved model parameters at a specified stage of training
Manifest	An inventory connecting named artifacts with versions, roles and integrity information
Replay	Repeating a recorded procedure to check whether its outputs agree

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Read each checkpoint against the same validation case

CHECKPOINT WEIGHT	TRAINING MSE	VALIDATION MSE
Update 1: a = 1.5	0.625	2.25
Update 2: a = 1.75	0.15625	0.5625

The baseline validation MSE is 9. The predeclared rule selects update 2 with validation MSE 0.5625 on one validation case. This is not a final-test score.

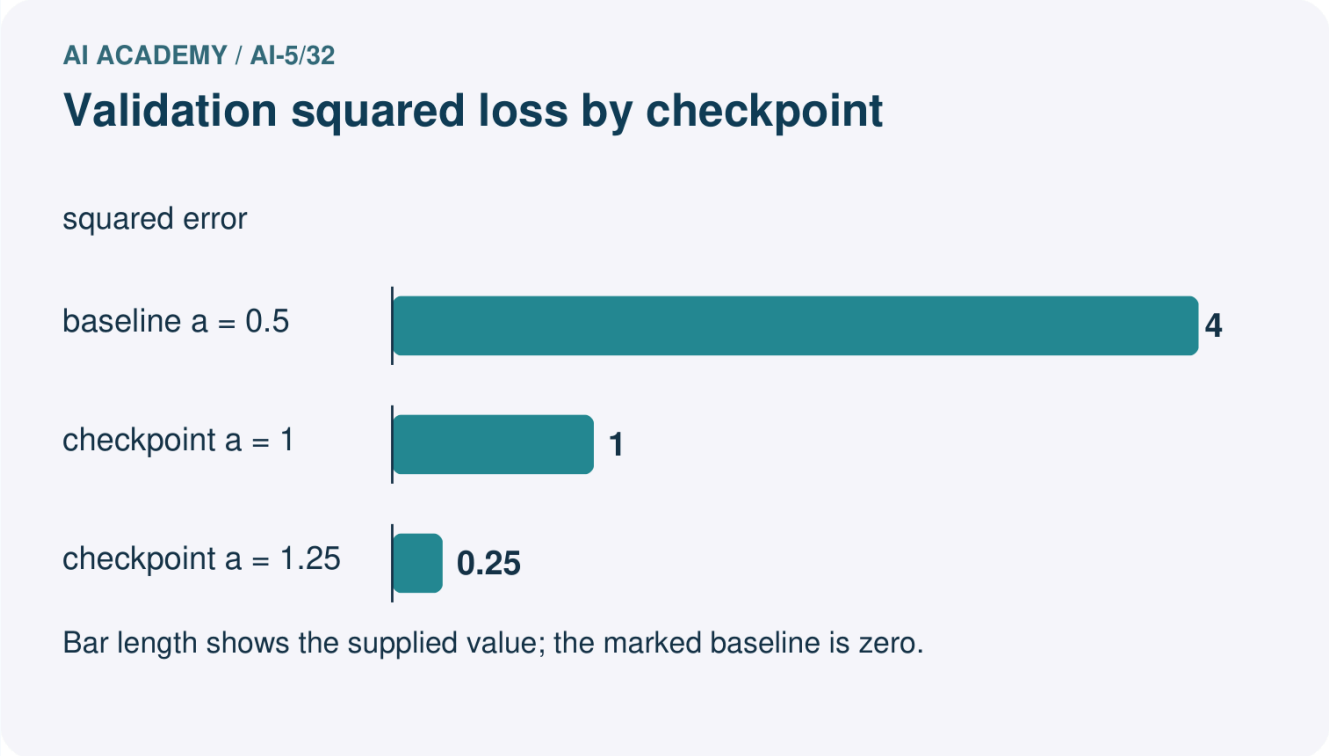
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Documenting a network experiment

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/32

Documenting a network experiment

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Baseline prediction

At a = 0.5, prediction 1, error2 = 4.



2. Checkpoint comparison

At a = 1, prediction 2, loss 1; at 1.25, prediction 2.5, loss 0.25.



3. Select and record

Choose a = 1.25; save R8, inputs, formula, checkpoint values, loss rule and validation role.

CHANGE ONE THING

A teammate calls V4 a final test after using it to select a=1.25.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Calculate both weight updates and give each **training** and validation MSE. Name the selected **checkpoint** and baseline validation MSE. How many independent validation examples support the comparison?

Teacher check: Teacher explanation: Weights are 1.5 then 1.75. **Training** MSE values are 0.625 and 0.15625; validation MSE values are 2.25 and 0.5625. Update 2 is selected; baseline validation MSE is 9.

There is one validation example, not two examples created by evaluating two **checkpoint**s.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Documenting a network experiment. Proposed explanation: “A smaller MSE is an accuracy percentage”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: MSE is an average squared error in squared target units; it is not classification accuracy or a percentage.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

What additional **evidence** is needed for a final performance claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Freeze the selected **checkpoint**, then evaluate an independently reserved final set; retain run settings, baseline and selection history. Changed-case check: Correct its role to validation; it was used for selection.

Changing the label on a dataset does not make its **evidence** untouched.

Changed-case reasoning: Correct its role to validation; it was used for selection. Changing the label on a dataset does not make its **evidence** untouched.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a record frame with question, fixed settings, data roles, raw results, selection, limitations and next action. Allow a calculator and compare **prediction**s before squared errors.

Extension: rename a data file without changing its bytes, then edit a value; explain why only the content change alters its content hash. Use copies of fictional classroom data.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the missing details in a vague performance claim and introduce run NN32-A.
7 to 18 minutes	Read the data roles and configuration. Complete W1 with a replay settings record.
18 to 32 minutes	Trace the two gradient updates, optionally run the code, and complete W2.
32 to 45 minutes	Use the visual to connect parameters, predictions, metrics and the selection decision.
Session 2 start to 8 minutes	Complete the guided V1 calculation and distinguish replay from independent evidence.
8 to 20 minutes	Build an artifact manifest and complete W3, including the limits of hashes and seeds.
20 to 35 minutes	Collect W4 independently, requiring raw predictions, errors and a bounded validation claim.
35 to 45 minutes	Use W5 and the exit response to repair records. Give feedback, then administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main weights 1.5/1.75, train MSE .625/.15625, validation 2.25/.5625, baseline validation 9. Select update 2 using one validation case. Independent R17 V9 x4 target 8: weights 1/1.5/1.75 give predictions 4/6/7 and squared errors 16/4/1; select 1.75, validation MSE 1 on one selected-on case. Final performance and runtime not measured.

Preserve exact versions and original artifacts; hashes detect change without certifying correctness.

Diagnose the current misconceptions

A smaller MSE is an accuracy percentage: MSE is an average squared error in squared target units; it is not classification accuracy or a percentage.

A **replay** adds new evaluation **evidence**: Reusing the same rows checks procedure consistency, not performance on new independent cases.

A seed or hash proves the experiment is sound: Seeds and hashes help track execution or integrity; they cannot repair bad labels, leakage or an unsupported claim.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A separate run R17 records baseline a = 1 and **checkpoint**s a = 1.5 and a = 1.75. It uses **prediction** max(0, a*x) and one validation case V9 with x = 4, y = 8.

Compute baseline and **checkpoint prediction**s and squared errors, choose by lowest validation MSE, and write a bounded conclusion. Final evaluation and runtime have not been measured.

Teacher explanation: **Prediction**s are 4, 6 and 7; squared errors are 16, 4 and 1. Choose a = 1.75. In R17, that **checkpoint** lowers MSE on V9 from 16 to 1 relative to the baseline, but V9 also selects it.

This is one-case validation **evidence**; final performance and runtime are not measured.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Write the replay settings

For NN32-A, record the **prediction** rule, initial weight, bias policy, learning rate, update count, loss, **training**/validation IDs, data order or shuffling policy, and **checkpoint** selection rule.

Name one environment field still to record when you run it.

Teacher explanation: Predict max(0, a*x); start a = 1 with no bias; rate 0.1; two full-batch updates; MSE is mean squared **prediction** error. T1 and T2 train, V1 validates; no shuffling or normalisation.

Select lower validation MSE, ties to earlier update. Record the actual Python version, alongside the code and Python float arithmetic.

W2 Trace the selected result

Calculate both weight updates and give each **training** and validation MSE. Name the selected **checkpoint** and baseline validation MSE. How many independent validation examples support the comparison?

Teacher explanation: Weights are 1.5 then 1.75. **Training** MSE values are 0.625 and 0.15625; validation MSE values are 2.25 and 0.5625. Update 2 is selected; baseline validation MSE is 9.

There is one validation example, not two examples created by evaluating two **checkpoint**s.

W3 Audit the artifact record

A folder contains only best-model.json and a screenshot of 0.5625. Identify four missing artifacts or records and explain what a content hash can and cannot verify.

Teacher explanation: Missing items include exact data with IDs and splits, code/configuration, earlier **checkpoint**s, raw **prediction**s and metric definitions, selection rule, and environment details.

A hash can detect changes relative to recorded content; it cannot establish correct labels, sound splits or a valid conclusion.

W5 Investigate a replay mismatch

A partner gets different results after changing pre**process**ing and replacing the validation file. They keep the old run ID and overwrite the old score.

Explain what should be preserved and how the revised work should be recorded.

Teacher explanation: Preserve the original data, pre**process**ing, settings, **output**s and run ID. Compare exact artifact versions to identify the mismatch.

Record changed pre**process**ing and data under a new run or explicit version, including reasons and results. A changed validation set is a different comparison; do not overwrite it as though it were the same run.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: 1.75*3 = 5.25, error 5.25 - 6 = -0.75, squared error 0.5625. V1 selected the **checkpoint** and is validation, not untouched final evaluation.

Exit: replace final accuracy with validation MSE 0.5625 on one constructed case selected using that case. Energy is not measured.

Supply data and splits, exact code, settings, baseline and both **checkpoint**s, raw **prediction**s, metric definition, selection rule and environment information.

W1 Write the replay settings

W2 Trace the selected result

W3 Audit the artifact record

W4 Audit a new record independently

A separate run R17 records baseline a = 1 and **checkpoint**s a = 1.5 and a = 1.75. It uses **prediction** max(0, a*x) and one validation case V9 with x = 4, y = 8.

Compute baseline and **checkpoint prediction**s and squared errors, choose by lowest validation MSE, and write a bounded conclusion. Final evaluation and runtime have not been measured.

Teacher explanation: **Prediction**s are 4, 6 and 7; squared errors are 16, 4 and 1. Choose a = 1.75. In R17, that **checkpoint** lowers MSE on V9 from 16 to 1 relative to the baseline, but V9 also selects it.

This is one-case validation **evidence**; final performance and runtime are not measured.

W5 Investigate a replay mismatch

Independent assessment and fresh retry

W4 requires **prediction**s 4, 6, 7 and squared errors 16, 4, 1; choose a = 1.75 and label the result one-case validation with final evaluation and runtime not measured.

Fresh retry: R18 has the same baseline and **checkpoint**s but validation x = 5, y = 10. **Prediction**s are 5, 7.5, 8.75; squared errors are 25, 6.25, 1.5625. Choose a = 1.75.

Require a validation-only conclusion and a record of which validation data version was used.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled documenting a network experiment **evidence** sheet to Image-pattern Network Investigation. Show how you can record a network experiment with enough detail to **replay** its calculations.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a record frame with question, fixed settings, data roles, raw results, selection, limitations and next action. Allow a calculator and compare **prediction**s before squared errors.

Extension: rename a data file without changing its bytes, then edit a value; explain why only the content change alters its content hash. Use copies of fictional classroom data.

Week 33 starts the capstone by recording a hypothesis, baseline, evaluation plan and success criteria before running controlled experiments.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/33, ai-5/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Record a network experiment with enough detail to **replay** its calculations

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Calculate both weight updates and give each **training** and validation MSE. Name the selected **checkpoint** and baseline validation MSE.

How many independent validation examples support the comparison?

Reasoned practice answer: Teacher explanation: Weights are 1.5 then 1.75. **Training** MSE values are 0.625 and 0.15625; validation MSE values are 2.25 and 0.5625. Update 2 is selected; baseline validation MSE is 9.

There is one validation example, not two examples created by evaluating two **checkpoint**s.

Check every learner before discussion: Use the worked case for Documenting a network experiment. Proposed explanation: “A smaller MSE is an accuracy percentage”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: MSE is an average squared error in squared target units; it is not classification accuracy or a percentage.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A teammate calls V4 a final test after using it to select a=1.25. Require a **prediction** and a reason before comparing. Expected result: Correct its role to validation; it was used for selection.

Reason: Changing the label on a dataset does not make its **evidence** untouched.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project **checkpoint**: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

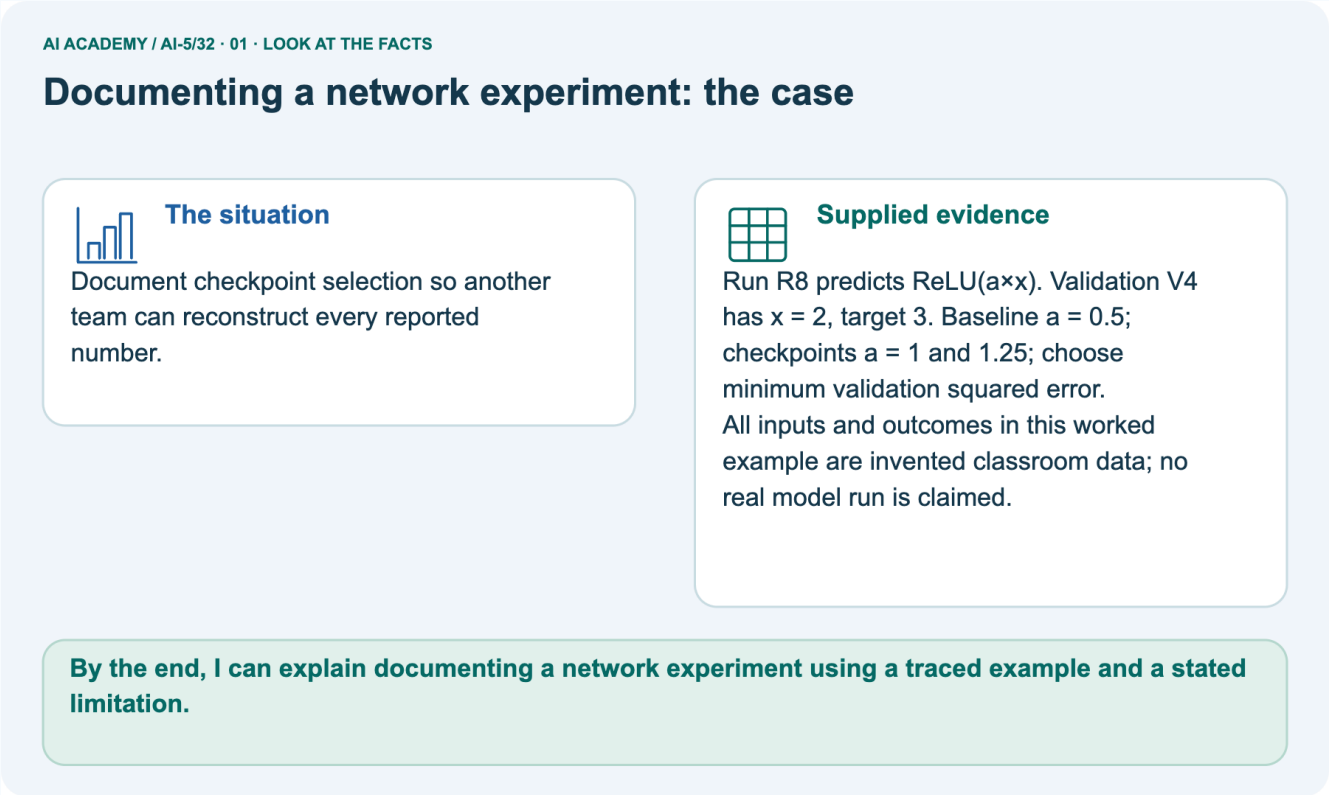
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Documenting a network experiment: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

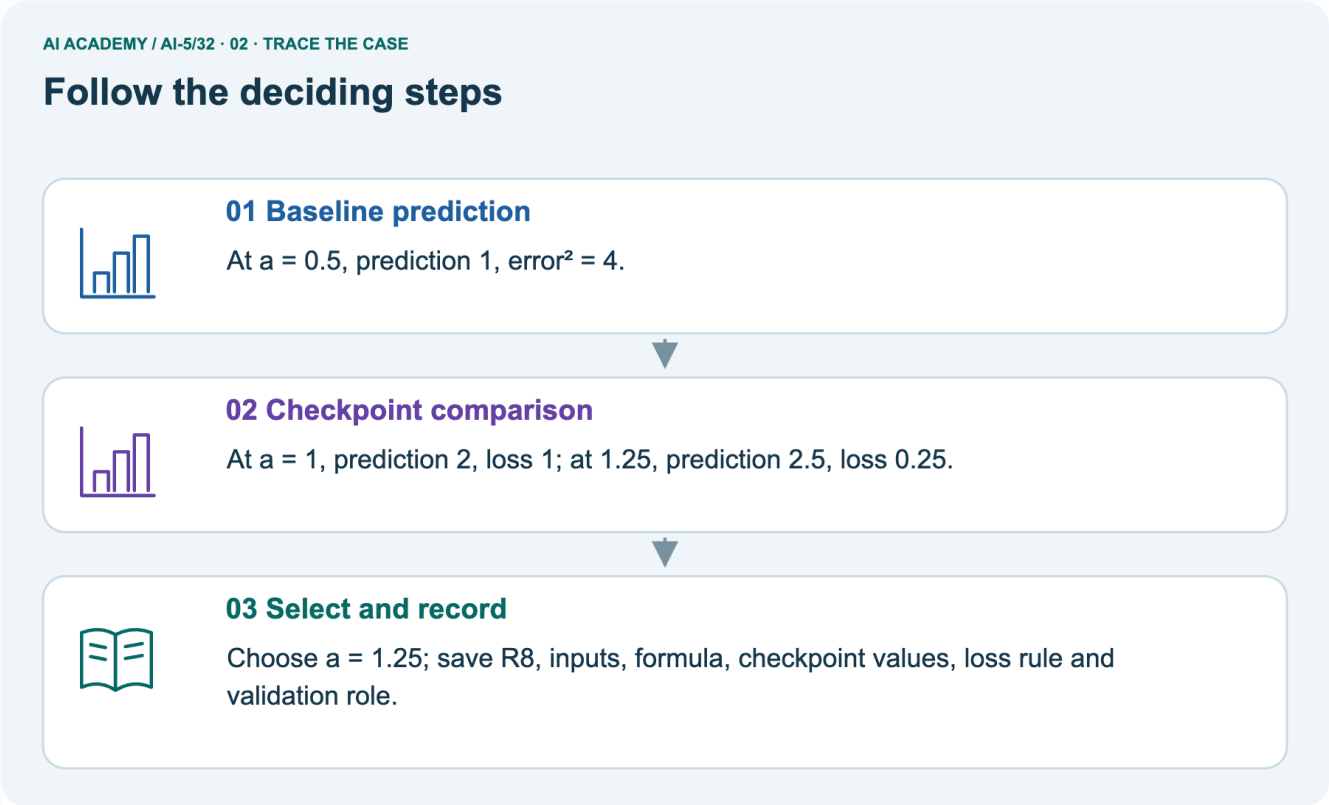
Document **checkpoint** selection so another team can reconstruct every reported number. Supplied facts: Run R8 predicts ReLU(a×x). Validation V4 has x = 2, target 3.

Baseline a = 0.5; **checkpoint**s a = 1 and 1.25; choose minimum validation squared error. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



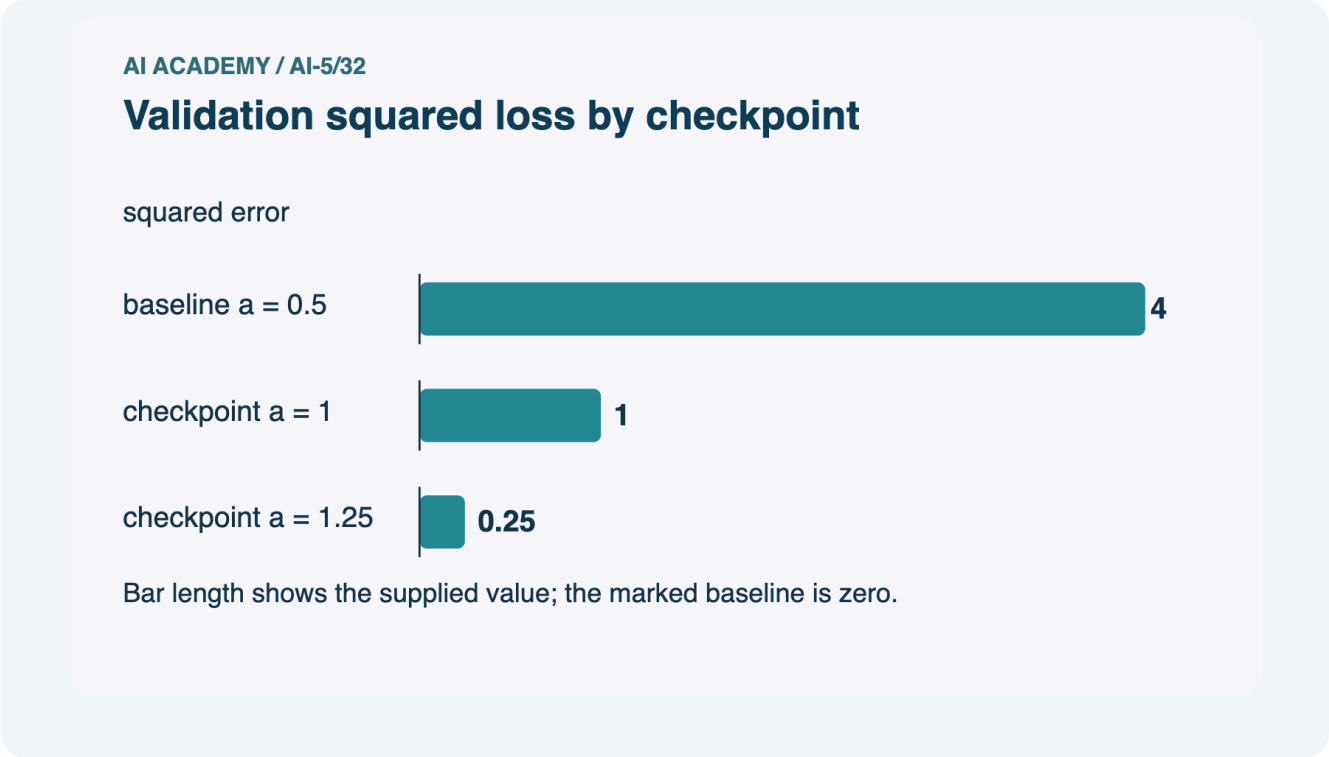
Follow the labelled picture with the matching explanation. Open the vector diagram

Baseline **prediction**: At a = 0.5, **prediction** 1, error² = 4. → **Checkpoint** comparison: At a = 1, **prediction** 2, loss 1; at 1.25, **prediction** 2.5, loss 0.25. → Select and record: Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Validation squared loss by checkpoint



Follow the labelled picture with the matching explanation. Open the vector diagram

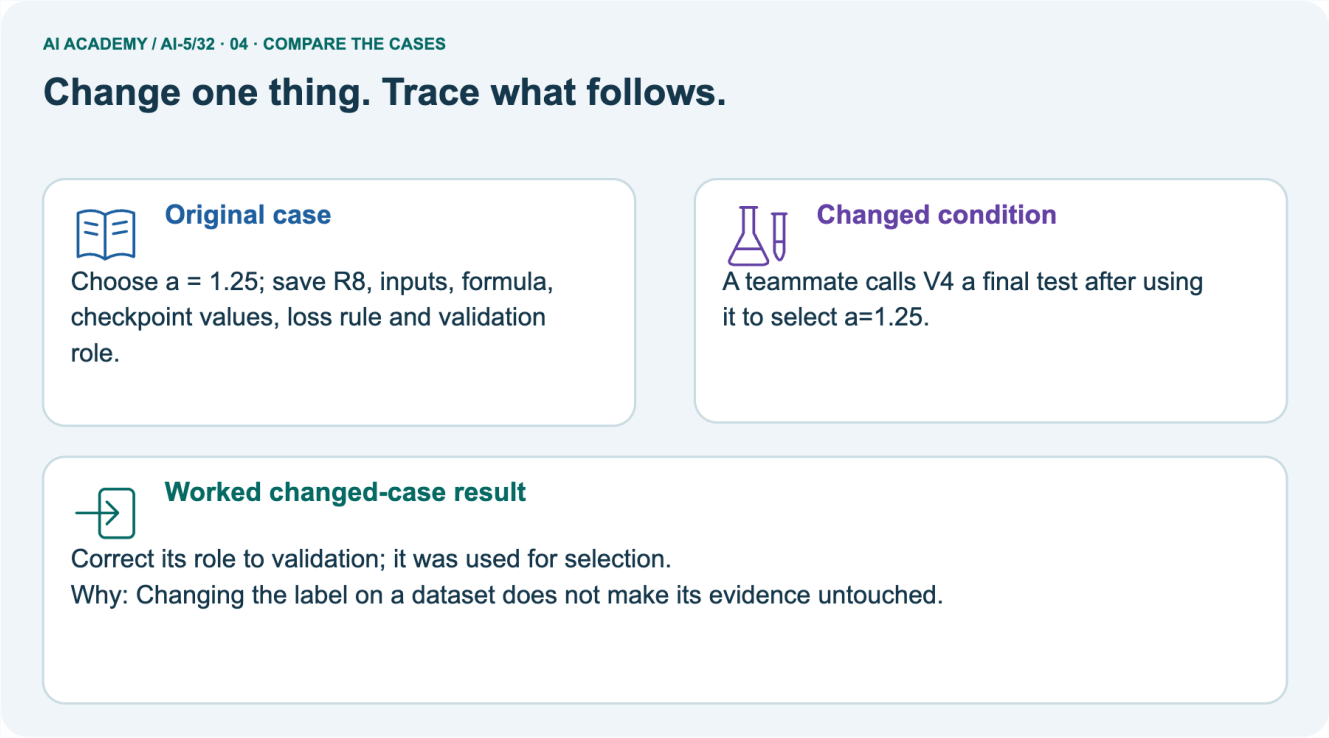
Validation squared loss by **checkpoint**. Units: squared error. baseline a = 0.5: 4. **checkpoint** a = 1: 1. **checkpoint** a = 1.25: 0.25.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Choose a = 1.25; save R8, **input**s, formula, **checkpoint** values, loss rule and validation role. Changed condition: A teammate calls V4 a final test after using it to select a=1.25.

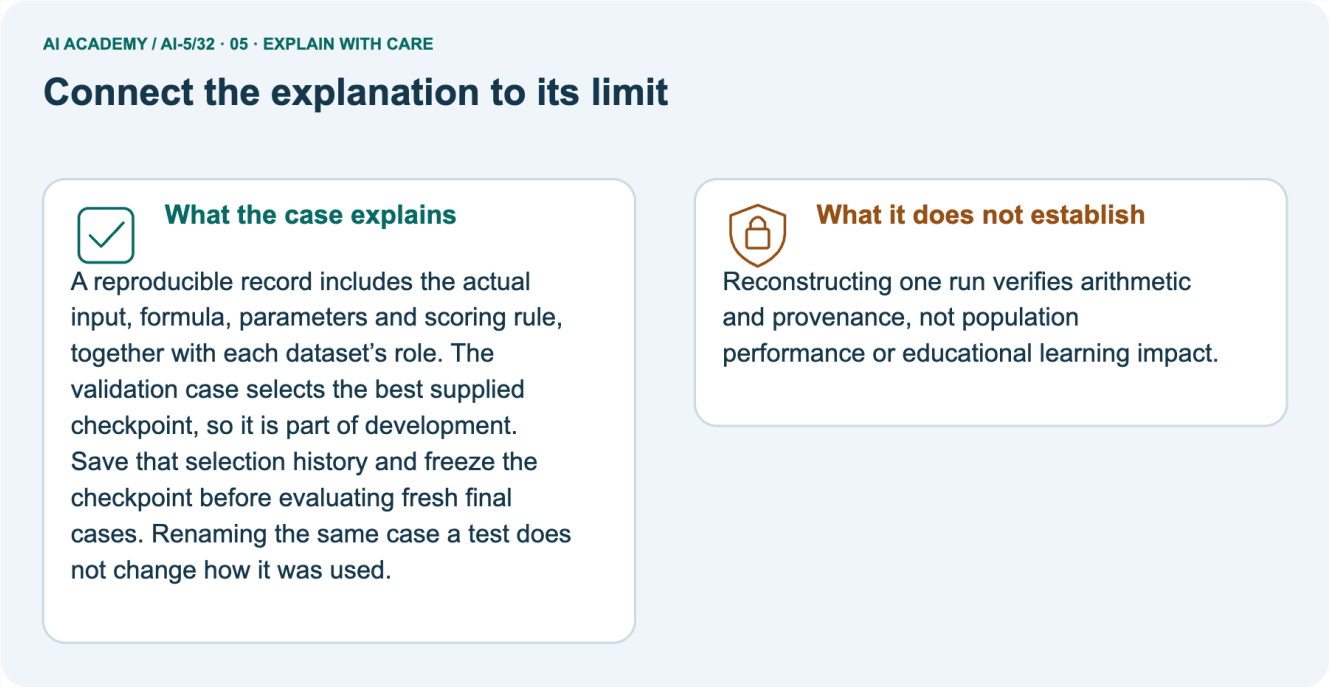
Expected result: Correct its role to validation; it was used for selection. Reason: Changing the label on a dataset does not make its **evidence** untouched.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Changing the label on a dataset does not make its **evidence** untouched.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A reproducible record includes the actual **input**, formula, parameters and scoring rule, together with each dataset's role. The validation case selects the best supplied **checkpoint**, so it is part of development.

Save that selection history and freeze the **checkpoint** before evaluating fresh final cases. Renaming the same case a test does not change how it was used.

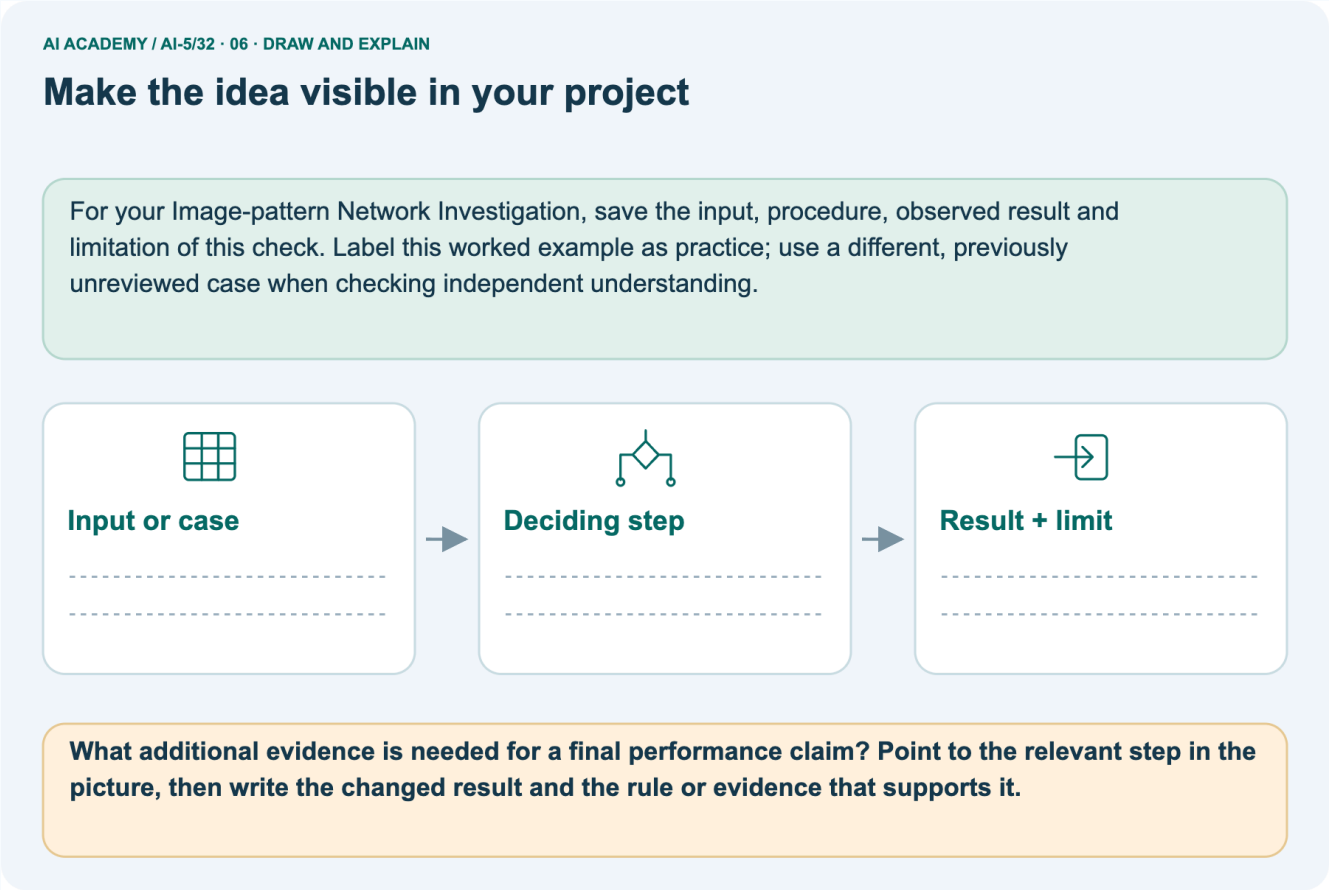
Limit: Reconstructing one run verifies arithmetic and provenance, not population performance or educational learning impact.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Reconstructing one run verifies arithmetic and provenance, not population performance or educational learning impact.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What additional **evidence** is needed for a final performance claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Freeze the selected **checkpoint**, then evaluate an independently reserved final set; retain run settings, baseline and selection history.

Changed-case check: Correct its role to validation; it was used for selection. Changing the label on a dataset does not make its **evidence** untouched.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Documenting a network experiment: the case

02. Follow the deciding steps

03. Validation squared loss by checkpoint

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.